

3rd Event-driven Business Process Management Workshop (edBPM)

Co-Chairs: Young Yoon and Hans-Arno Jacobsen, University of Toronto;
Allen Chan and Phil Coulthard, IBM Canada Ltd.;
Adrian Paschke, Freie Universität Berlin

Theme: SOA 2.0 & Smart Services

Many companies are reporting astronomical increases in event activities in enterprise IT systems worldwide. IBM reports that 72 quadrillion unique business events are generated worldwide each day. Although it is still a daunting task to design and develop systems that are capable of handling this amount of events, this trend constitutes an opportunity to drive the study of efficient, fast, and reliable BPM solutions that give rise to the processing of events at that scale. Gartner's recent report projects that market for business intelligence (BI) and business activity monitoring (BAM) will have a compound annual growth rate of 7 to 8% by 2012, and that platforms for enriched event management and processing will be in high demand.

Early workflow processing systems viewed a process as a sequential handoff of information from task to task, with each task responsible for retrieving, transforming, and updating data to and from external applications. In today's service-oriented world, such a model is too slow, too difficult to implement, too inflexible to adapt and too costly to maintain. Business Process Management (BPM) is a more agile and flexible model that views process activities as actions triggered automatically in response to the events generated in disparate systems throughout the enterprise. An event is an asynchronous state transition that signals the change of internal or external data, such as the update of an address, the addition of contact information and the arrival or cancellation of an order. Most applications already generate events, if only for their own internal needs. Today, BPM adds a common middleware framework for detecting events, communicating them to an integration hub, converting them to a common format, and linking them to triggered actions in a given business process.

The recently coined term “Event-Driven Business Process Management” (EDBPM) refers to an enhancement of traditional BPM through novel combinations of Service Oriented Architectures (SOA), Event Driven Architectures (EDA), Software as a Service (SaaS), Business Activity Monitoring (BAM) and the Processing of Complex Events (CEP). BPM in turn, refers to a software platform which provides companies and end-users the ability to model, manage and optimize business processes.

Loosely coupled event-driven architectures for BPM provide the following important benefits:

- **Responsiveness:** Events can occur at any time from any source; processes respond to them immediately, whenever and wherever they happen.
- **Agility:** New processes can be modeled, implemented, deployed and optimized more quickly in response to changing business requirements, often in a completely dynamic fashion.
- **Flexibility:** Processes can span heterogeneous platforms and programming languages; participating applications can be upgraded or changed without breaking the process model.

This workshop introduced the state-of-art concepts underlying EDBPM solutions, summarized standard best practices in event management, and discussed the shortcomings, challenges and obstacles. Specifically, there has been little understanding of service impact beyond event severity and root-cause of business events. This brought attention to more sophisticated correlation mechanisms and infrastructure to reason about dependency between events and their sources. Also, we discussed the various methods of mapping event incidents to customer SLA and KPI that are crucial for identifying any escalating problems critical to the business.

Given the motivational challenges of aligning events with business process and services, the workshop hosted a series of presentations about visionary and practical solutions: (1) the implication of exposing events useful for BPM from the service interface and issues with implementation; (2) noble SLA modeling on dynamic composition of business processes; (3) automated evolution of business process upon detection and resolution of semantically conflicting business events; and (4) universal and highly expressive complex event processing language that can be effective in deriving knowledge.

The main goal of the workshop was to create awareness for the role of event processing within BPM, define the challenges and start establishing a research community around these areas. The workshop covered the following topics:

- Event-driven BPM: Concepts
- Role of event processing in BPM
 - Business Events: types and representation
 - Event stream processing in business processes
- Data- and event-driven business processes
 - Evaluation and ROI of event-driven BPM
 - Event-driven SOA
 - EDA and BPM
 - Real-time awareness in BPM
 - Context in BPM
- Event-driven BPM: Patterns
- Definition, Creation, Representation, and Learning
 - BPMN and event processing
 - Modeling unknown and similar events in business processes
 - Modeling events in human-oriented tasks
 - Semantics and ontologies for event-driven BPM
 - Publish/Subscribe mechanism and process modeling
- Runtime CEP and BPM
 - Event pattern detection
 - BPEL and event processing
 - Reasoning about unknown/similar events
 - Distributed event processing
 - Dynamic workflows
 - Ad-hoc workflows
- Applications and use cases for event-driven BPM
 - Event-driven monitoring and BAM
 - Event-driven SLA monitoring